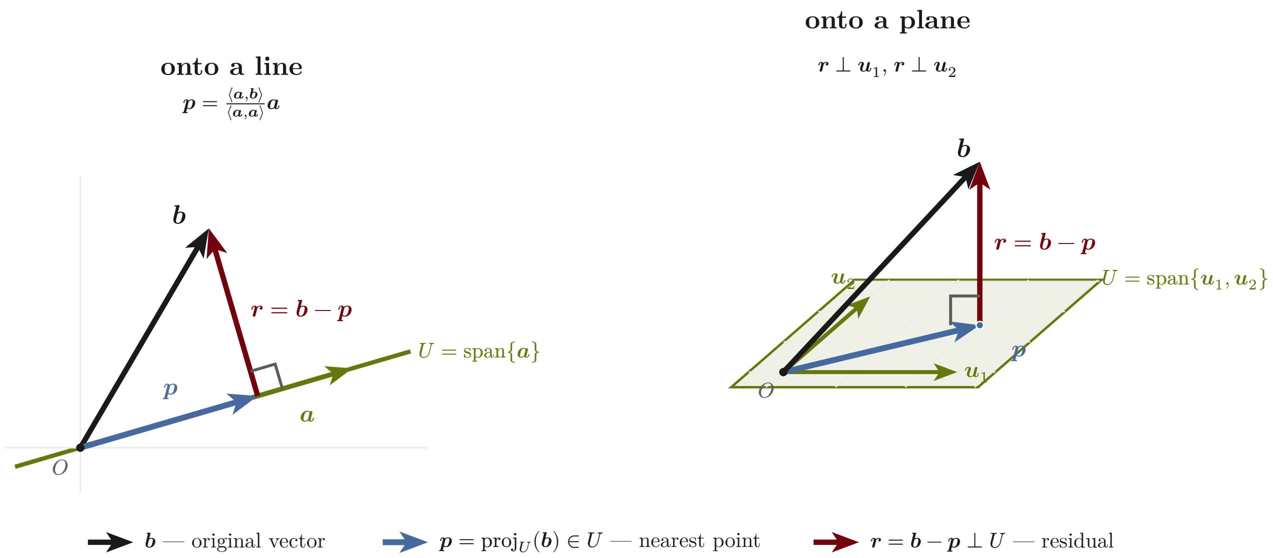
# Orthogonal projection onto a subspace




defining property: the residual is orthogonal to the whole subspace, so $\boldsymbol{p}$ minimizes $\|\boldsymbol{b} - \boldsymbol{x}\|$ over $\boldsymbol{x} \in U$ — the basis of least squares and PCA.